

# UNIT - 3 :

- **System Modeling : Definition**
  - Types
  - Categories :
    - Structural Models
    - Behavioral Models
    - Interaction Models
  - Examples
- **Structural Models :**
  - **Introduction**
  - **Purpose**
  - UML
  - Entities
  - Classes
  - **Relationships :**
    - Dependency
    - Association
    - Aggregation
    - Composition
    - Generalization/Inheritance
    - Multiplicity
  - **Examples / Question**
- **Behavioral Models :**
  - State Model - Introduction
  - **Event vs State**
  - Transitions and Conditions
  - Stage Diagram
    - Components
    - Continuous Loops diagrams
    - One-Shot Lifecycle diagrams
    - **Examples / Question**
- **Interaction Models :**
  - Sequence Models - Introduction
  - 2 Kinds
  - **Sequence Diagram**
    - Components
    - Types of Messages
    - **Examples / Question**
  - Activity Models - Introduction
  - Activity Diagram
    - Components
    - **Examples / Question**

## UNIT - 4 :

- Software Project Management (SPM) - Introduction
  - Phases
  - 4 Important Tasks in SPM
  - Activities covered by SPM
- Project Manager
  - What ? Who ?
  - Skills
  - Responsibilities
- Project Management
  - Introduction
  - Activities - 4 Stages
  - Reasons for failure
- Project :
  - What ?
  - Characteristics
  - Difference with Operations
  - Difference with Jobs
  - Software Project Vs other types of project

- Project Evaluation - Introduction
  - CBA
  - Cash Flow Forecasting
- Cash Benefit Evaluation Techniques :

1. Net Profit :
  - a. Definition
  - b. Formula
  - c. Example / Problem
2. Payback Period (PbP) :
  - a. Definition
  - b. Formula
  - c. Example / Problem
3. Return on Investment (ROI) :
  - a. Definition
  - b. Formula
  - c. Example / Problem
4. Net Present Value (NPV) :
  - a. Definition
  - b. Theory (*in slides*)
  - c. Formula
  - d. Example / Problem
5. Internal Rate of Return (IRR) :
  - a. Definition
  - b. Theory
  - c. Formula
  - d. Steps
  - e. Example / Problem